HCLTech | AI Future Ready FullStack Web Engineer

4. SKILL GAP AND WORKFORCE EVOLUTION PREDICTOR

SkillPulse AI — Workforce Intelligence Platform

Skills Knowledge Graph • Vector RAG • Scikit-Learn Forecasting • 5 AI Agents • Explainable AI

Squad Name – Arsenal

Role – AI_Full Stack Web Engineer

Team:

- **Mohit Tomar – HCLTFP2460368**
- **Ronit Singh Rawat – HCLTFP2465203**
- **Ankit Mohapatra – HCLTFP2467657**

Use Case Overview

- Use Case: 4. Skill Gap & Workforce Evolution
- Industry: HR / Workforce Transformation
- Goal: Predict skill shortages 6–24 months early
- Output: Personalized courses & internal mobility
- Safety: 0% Demographic Bias + HR Approval

1. Real-World Problem, Inputs & Our Solution

Simple view of what problem we solve and what data enters the system

The Real-World Problem

1. Late Skill Discovery:

Companies find skill gaps AFTER winning projects.

2. Expensive External Hiring:

Hiring outside contractors costs 3x more money.

3. Hidden Internal Talent:

Keyword search misses people with 80% close skills.

The 5 Inputs to Our Project

1. Employee Skills: 1–5 sliders or Resume text.

2. Learning Records: Past certifications & speed.

3. Project Requirements: Target role or project spec.

4. Market Trends: Gartner/WEF tech growth reports.

5. Tech Evolution: Emerging, Core & Legacy skills.

How SkillPulse AI Solves It (Business Impact)

- **Predicts Shortages Early: Forecasts skill demand 6, 12, and 24 months ahead (93.1% accuracy).**
- **Finds Adjacent Talent: Uses a Skills Knowledge Graph to find engineers who can upskill in 4 weeks.**
- **Saves 87% Hiring Cost: Reskills internal staff instead of expensive external recruiting.**
- **100% Fair & Human-Controlled: Strips Age/Gender bias (0% weight) and requires HR approval.**

2. High-Level System Architecture & Tech Stack

5 simple layers from React UI to FastAPI + Pydantic, 5 AI Agents, and Databases

LAYER 1: FRONTEND UI (React 18 + TypeScript + Tailwind CSS + Zustand)

Interactive Skill Sliders • What-If Forecast Sliders • Visual Knowledge Graph • HR Approval Screen

LAYER 2: API & VALIDATION GATEWAY (Python FastAPI + Pydantic + JWT Security)

Pydantic validates all inputs/outputs (1–5 ratings, JSON schemas) • Bias Filter removes Age/Gender • JWT Auth

LAYER 3: 5-AGENT COORDINATOR (LangGraph Stateful Workflow)

Agent 1: Profiler → Agent 2: Forecaster → Agent 3: Learning Coach → Agent 4: Mobility → Agent 5: XAI Auditor

LAYER 4: AI & ML BRAIN (Skills Knowledge Graph + Scikit-Learn + Hybrid LLM)

Knowledge Graph (45 Skills, 37 Bridges) • Scikit-Learn Ridge Forecaster • Local + Cloud Hybrid LLM

LAYER 5: STORAGE & VECTOR RAG (SQLite Database + FAISS-Style Vector Store)

SQLite (120 Employees, 1,080 Time-Series Rows, Audit Logs) • Vector Store (40+ Market & Course Docs)

3. Why This Tech Stack (And What Pydantic Does)

Why we chose each tool from the brief over other alternatives

Why React + FastAPI + Pydantic?

- **React + TypeScript + Tailwind + Zustand:**
  3x faster UI & SVG graph rendering than Material UI/Redux.
- **Python FastAPI (Why Not Java or Node.js?):**
  Python directly runs Scikit-Learn, Vector RAG & AI Agents.
- **What We Do With Pydantic:**
  Strictly validates API data (e.g. skill levels 1–5, horizon 6–24m) and auto-creates OpenAPI (/docs) schemas.

Why Scikit-Learn + Local Vector RAG + SQLite?

- **Scikit-Learn + Hybrid LLM (Why Not LLM Alone?):**
  LLMs are bad at math. Scikit-Learn gives accurate forecasts (6.8% error) and works 100% offline without API failures.
- **Local Vector RAG (Why Not Cloud Search?):**
  Searches courses & market reports in <3ms with 0 cloud keys.
- **SQLite (Why Not Heavy DB Servers?):**
  Zero setup; runs pre-seeded with 120 profiles in 1 command.

How Pydantic Protects Our AI Pipeline (4 Key Jobs in Code)

1. **Input Validation:** Checks that skill sliders (0–5), horizons (6–24m), and role IDs are valid before AI runs.
2. **HR Governance Rules:** Enforces pattern='^(APPROVED|REJECTED)$' so invalid HR statuses are blocked.
3. **Clean JSON Serialization:** Guarantees XAI factor weights and 5-Agent traces match the frontend TypeScript types.
4. **Auto Swagger UI (/docs):** Generates interactive API test forms automatically from our Pydantic BaseModel classes.

4. End-to-End Working Flow (API & Data Flow)

When each API is called and where data is fetched from

STEP 0: Server Startup

Loads 120 employees & 24m history from SQLite → Builds Skills Graph in RAM → Trains Scikit-Learn Forecaster.

FLOW 1: User Runs 5-Agent Simulator (POST /api/v1/agents/simulate-custom-profile)

Pydantic validates input → Bias Filter strips Age/Gender → 5 Agents query Graph + Vector RAG → Saves plan to SQLite.

FLOW 2: User Moves What-If Forecast Sliders (POST /api/v1/forecast/simulate)

Fetches 24m history from SQLite → Scikit-Learn predicts +6/12/24m demand & shortage risk → Updates UI charts.

FLOW 3 & 4: Skills Graph & Internal Mobility (GET /graph/skills-network & /mobility/match)

Fetches 45 skill nodes from Graph + screens 120 SQLite employees → Ranks best internal candidates & cost savings.

FLOW 5: Bias Test & Human HR Approval (POST /guardrail/test & /governance/decision)

Blocks biased Age/Gender queries → Verifies HR JWT token → Updates SQLite status to APPROVED + logs Audit Trail.

5. The 5 AI Agents, Balanced Dataset & Results

How the 5 agents divide the work and our verified accuracy metrics

What the 5 AI Agents Do

- **Agent 1 (Profiler): Finds skill gaps & graph bridges.**
- **Agent 2 (Forecaster): Checks RAG trends & ML demand.**
- **Agent 3 (Learning Coach): Picks courses & career path.**
- **Agent 4 (Mobility): Matches internal staff & $ saved.**
- **Agent 5 (XAI Auditor): Shows Why-% & proves 0% bias.**

Balanced Dataset & Evaluation Metrics

- **Balanced Synthetic Dataset:**

  120 Employees (6 depts) • 45 Skills • 1,080 History Rows.

- **Forecast Accuracy: 93.16% (6.84% MAPE, R^2 = 0.94).**
- **Course Match Precision: 94.8% (RAG Grounded).**
- **Demographic Bias Weight: 0.00% (Age/Gender stripped).**

All 10 Functional Requirements Fulfilled (FR-01 to FR-10)

- **FR-01: Skill Inventory (Sliders + Resume Parser)**
- **FR-02: Skill Gap Detection (Knowledge Graph)**
- **FR-03 & 04: Market RAG & 6/12/24m ML Forecast**
- **FR-05: Personalized Week-by-Week Courses**
- **FR-06: Internal Mobility (Rankings + Cost Saved)**
- **FR-07: Workforce Risk (Shortage & Bus-Factor Alerts)**
- **FR-08 & 09: 3-Step Career Plan & Explainable AI (XAI)**
- **FR-10: Human HR Approval Queue & Bias Refusal Guardrail**